

FATIMA ALBURAIKI

AI Engineer | Data Scientist | Software Developer | Technical Instructor

FatimaAlburaiki@outlook.sa | +966 50 672 0367 | Portfolio : fatimaalb.github.io

Github: github.com/FatiMaAlb | LinkedIn: linkedin.com/in/Fatimah-Alburaiki | googlescholar

About Me

I hold a Bachelor's degree in Computer Science with a specialization in Data Analysis and Artificial Intelligence. Experienced in machine learning, deep learning, data preprocessing, and building AI-driven and LLM-based applications. Skilled in developing predictive models, RAG systems, and data visualization dashboards. Seeking a role in AI or Data Science to apply technical expertise and contribute to impactful projects.

Technical Skills

- Programming: Python, SQL, Java, C#, JavaScript, Swift
- Data Analytics: Data Cleaning, EDA, Feature Engineering, Data Validation, Power BI
- ML & DL: Regression, Classification, CNNs, Hyperparameter Tuning
- AI & LLMs: OpenAI API, LangChain, RAG, Prompt Engineering, NLP
- Web Development: ASP.NET, FastAPI, React.js, HTML, CSS, XML
- Databases: PostgreSQL, SQL Server, MySQL, Firebase
- Cloud & DevOps: AWS, Docker
- Libraries & Tools: Pandas, NumPy, TensorFlow, Scikit-learn, Looker Studio, n8n
- Operating Systems: Windows, Linux

Experience

Technical Coordinator – AI & LLM Solutions

Sep 2025 - Present

Policy Generation Funded Research Project, **National Cybersecurity Authority**

- Coordinated the development workflow of an LLM-based AI project from inception to implementation.
- Guided technical decisions and provided expertise throughout the project lifecycle.
- Oversaw data collection, cleaning, validation, and structuring processes.
- Utilized large language models for policy analysis, information extraction, and knowledge organization.
- Identified and resolved technical challenges to ensure project progress and quality.

Digital Skills Teacher, Al-Rowad & Faheem Platform

Sep 2025 - Present

- Teaching digital skills to middle school students. I explain various technical concepts, conduct scientific labs, and supervise students' technical projects at school.

Website development, Imam Abdulrahman University

Jul 2025 - Aug 2025

- Worked as a developer, handling programming and platform-related tasks for the Tawtheeq platform and the Teaching Assistants & Lecturers platform.

Content Developer and Instructor, SkillX

Jan 2024 - April 2025

- Designed and delivered comprehensive Power BI training on data analysis and visualization ,and Created course materials, exercises, and practical examples

Private Tutor (University-Level)

Sep 2023 - Mar 2024

- Provided one-on-one tutoring in Advanced **Database Systems, Human-Computer Interaction, Distributed and Parallel Systems, and Multimedia** .

Projects

Generative AI System for Cybersecurity Policy Generation – Research & Application Jun 2025 – Present

- Coordinated the development of an AI-driven cybersecurity policy generation system using LLMs.
- Led data preparation, knowledge organization, and policy analysis processes.
- Contributed to RAG pipeline design and workflow automation using LLM technologies.
- Collaborated on integrating OpenAI APIs and AI components into the solution.

Tawtheeq Platform – Web Application Jul 2025 – Sep 2025

- Designed and managed database structures using SQL for storing event and participant data.
- Implemented automated certificate generation based on stored data and system inputs.
- Developed backend processes for data handling, validation, and retrieval.
- Integrated database operations with web interfaces for efficient user interaction.

Facial Emotion Recognition Using CNN Models – Research & Application Dec 2024 – Jun 2025

- Preprocessed and augmented datasets to improve data quality and balance.
- Developed CNN models using TensorFlow for emotion classification.
- Tuned hyperparameters and applied fine-tuning to improve performance.
- Evaluated models using accuracy, F1-score, and confusion matrix.

Business & Application Data Analysis Dashboards – Power BI & Looker Studio 2025

- Analyzed supermarket sales and Google Play data to uncover trends, performance, and user behavior.
- Built interactive dashboards to track KPIs (sales, profit, categories, and app metrics).
- Identified relationships between features and overall performance.
- Cleaned and transformed data using preprocessing and Power Query techniques.
- Created dynamic visualizations to support data-driven decision-making.

Mobile App: HeadRest – Application Oct 2022 – Jul 2023

- Developed a mobile application to track and analyze migraine patterns using user-generated data.
- Collected and processed user data to identify potential triggers and behavioral patterns.
- Applied machine learning techniques to predict migraine occurrences.
- Provided personalized insights and recommendations based on data analysis.

Education

Najran University, BS in Computer Science Sept 2019 - June 2023

Certifications & Course Training

- **Introductory Artificial Intelligence Program** 40 Hrs– KAUST Academy, 2026
- **Machine Learning Scientist Track** 85 Hrs– DataCamp, 2025
- **Natural Language Processing Track** 20 Hrs – DataCamp, 2025
- **Mathematics for Machine Learning and Data Science** 99 Hrs - DeepLearning.AI (Coursera), 2025
- **Supervised Machine Learning: Regression** 20 Hrs , IBM (Coursera), 2024
- **Exploratory Data Analysis for Machine Learning** 14 Hrs – IBM (Coursera), 2024
- **Improving Deep Neural Networks:** 23 Hrs, DeepLearning.AI (Coursera), 2024
- **Python for Data Science Specialization** 132 Hrs– University of Michigan (Coursera), 2024

Research Experience & Publications

- **AI-Based Cybersecurity Policy Generation** – Ongoing Research
- **Emotion Recognition Using CNN-Based Models** – Accepted for publication at *AIS 2026* (IEEE)
- **Cryptographic Algorithms for Securing IoT Devices: A Survey** – Published in IEEE Xplore, *NCCC 2022*
- **Emotion-Aware Smart Environments: Detection Technologies within IoT Systems: A Survey** – Under Review
- **HeadRest: Design and Development of a Mobile Application for Migraine Tracking and Prediction** – Under Review

Conferences Participation

- **The 2nd International Conference on Artificial Intelligence Systems - IEEE AIS 2026** , San Antonio, USA
May 2026
- **The 5th International Conference on Computing and Artificial Intelligence - ICMI**, KFUPM , Al-Ahsa, Saudi Arabia
April 2026
- **The 6th National Computing Colleges Conference**, IAU, Dammam, KSA
May 2024
- **The 5th National Conference of Saudi Computer Colleges**, Umm Al-Qura University, Mecca, KSA
Dec 2022

Languages

- Arabic
- English